

CEN-348 Artificial Intelligence Systems
Spring 2021
Final Exam
Duration: 100 minutes

Note: Write your name, surname and school number to the left top of your paper. Please write your answers **step by step**. **Draw a box** around your final results. Please check your answers **carefully**. Upload your answers as a **pdf** file. **Please do not cheat! Please do not share your answers with the others. In this exam, there are 4 questions !!!**

- 1) Suppose a genetic algorithm uses chromosomes of the form $x = abcdefgh$ with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of individual x be calculated as:

$$f(x) = (a + b) - (c + d) - (e + f) - (g + h)$$

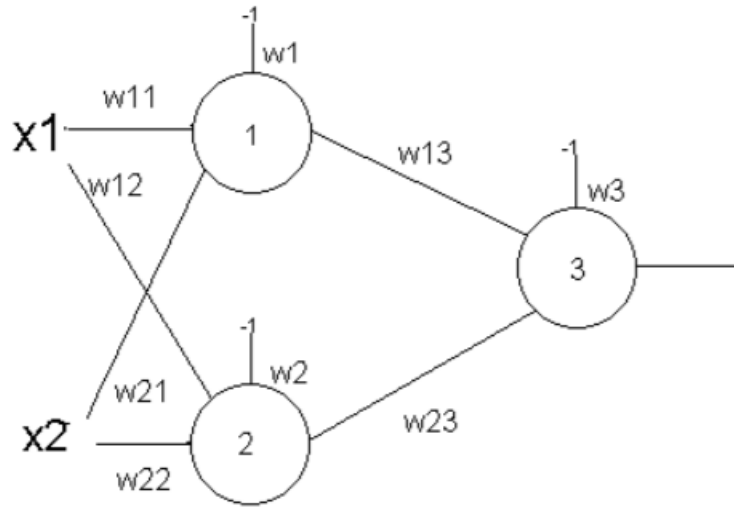
And let the population consist of five individuals with the following chromosomes:

$$\begin{aligned}x_1 &= 12345678 \\x_2 &= 23456781 \\x_3 &= 45637281 \\x_4 &= 53617284 \\x_5 &= 23921285\end{aligned}$$

- a) Compute the fitness value for each individual and sort them in order with the fittest first and the least fit last. (15 p)
- b) To create mating pool, roulette wheel selection is used. You will select the individuals. The random values to perform roulette wheel selection are 0.15, 0.56, 0.50, 0.68, and 0.95 respectively. Please construct the mating pool step by step and write the selected individuals clearly. (15 p)
- c) Perform the following crossover operations in the mating pool:
 - i) Cross the fittest two individuals using one-point crossover at the middle point. (5 p)
 - ii) Cross the second and third fittest individuals using a two-point crossover (position 3 and 7). (5 p)

- 2) The following simple neural network has 3 units labeled 1, 2, and 3. All units are negative sigmoid units, which means that their output is computed with the equation as follows: $n(z) = -\frac{1}{1+e^{-z}}$. The equation of the derivative of $n(z)$ is $\frac{dn(z)}{dz} = n(z)(1 + n(z))$.

This network uses the error function $E = \frac{1}{2}(2y^* - 2y)^2$.



The initial weights are provided below.

Weight	w1	w11	w12	w13	w2	w21	w22	w23	w3
Value	0.5	1	1	0.5	0.5	1	1	0.25	0.5

According to these assumptions,

- a) The input vector $[x1 \ x2] = [2 \ 0.5]$, compute the output at each neuron after forward propagation. **Please draw a box for final result of forward pass. Please write your answers step by step.** (10 p)

y1 (output at neuron 1)

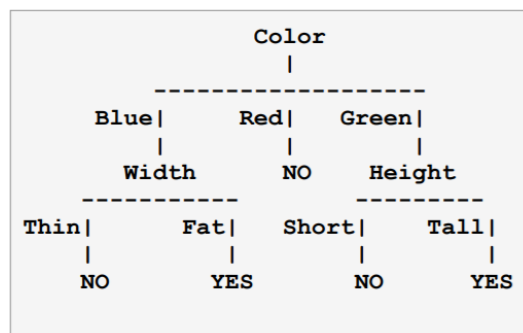
y2 (output at neuron 2)

y3 (output at neuron 3)

- b) Using a learning rate of 1, the input vector in (a), and a desired output of 0, backpropagate for the selected weights in the table below for only once. **Please write your answers step by step.** (15 p)

Weight	w_3	w_{13}	w_{11}
Updated value			

- 3) According to the given decision tree, indicate how the examples in the following table is classified by writing to the last column of the table. (15 p)



Example	Color	Height	Width	Class
1	Red	Short	Thin	
2	Blue	Tall	Fat	
3	Green	Short	Fat	
4	Green	Tall	Thin	
5	Blue	Short	Thin	

- 4) NASA would like to distinguish Martians (M) from Humans (H) by using the related characteristics: Green $\in \{N, Y\}$, Legs $\in \{2, 3\}$, Height $\in \{S, T\}$. Training data is given as follows:

Example	Green	Legs	Height	Species
1	N	3	S	M
2	Y	2	T	M
3	Y	3	T	M
4	Y	2	S	M
5	Y	3	T	M
6	N	2	T	H
7	N	2	T	H
8	N	2	T	H
9	Y	2	T	H
10	N	2	T	H

Build the ID3 tree by using the information gain splitting criterion. Please write your results step by step and construct the tree step by step. **If you don't write your computations, your built tree is invalid to take point.** (20 p)

Good Luck!